

Correction for Theorem 5.3 of MPR

1 Preliminaries

We adopt the notation from the main paper. Let $\mathcal{A}$ denote the dataset of size N , where each sample is $z_i = (u_i, v_i, s_i, r_i)$. $S \in \{0, 1\}$ denotes the true sensitive attribute, and $\hat{R}$ denotes the predicted rating.

We refine the original Assumption 4.1 to the following assumptions, while retaining the remaining problem setup and notations from the original paper. This modification aligns with the CTR prediction scenario and preserves the validity of other theoretical derivations in the original paper, as the DP metric relies solely on the linearity of expectation independent of the distribution shape.

Assumption 1. *The predicted ratings (or probabilities) are bounded. Without loss of generality, we assume $\hat{R} \in [0, 1]$.*

Assumption 2. *Let $\mu_0 = \mathbb{E}[\hat{R}|S = 0]$ and $\mu_1 = \mathbb{E}[\hat{R}|S = 1]$. We assume that the expected ratings of the two groups are not consistent, i.e., $\mu_0 \neq \mu_1$.*

2 Main Theorems

We present two theorems establishing the generalization error bounds for MPR. Theorem 1 assumes the true prior exists in the candidate set $\mathcal{P}$.

Theorem 1 (Generalization Bound). *Assume the true distribution $p_0 \in \mathcal{P}$. Let the minimum group size in any valid dataset be bounded by $N_{\min} \geq \lceil \delta N \rceil$ for a safety threshold $\delta \in (0, 0.5)$. For any $\eta \in (0, 1)$, with probability at least $1 - \eta$, the true fairness gap satisfies:*

$$|F| \leq L_F + \mathcal{M}(N, \eta, \delta) + C_\epsilon(p_0), \quad (1)$$

where L_F is the robust loss, and the terms are defined as:

$$C_\epsilon(p_0) = \frac{\epsilon}{p_0(1-p_0)}, \mathcal{M}(N, \eta, \delta) = \left(\frac{1}{\lceil \delta N \rceil - 1} + \frac{1}{N - \lceil \delta N \rceil + 1} \right) \sqrt{\frac{N \ln(|\mathcal{P}|/\eta)}{2}}. \quad (2)$$

3 Proofs

3.1 Reconstruction Error

We first quantify the fairness estimation error introduced by reconstruction errors ϵ . Using the Law of Total Probability and the linearity of expectation:

$$\mathbb{E}[\hat{R}|\hat{S} = 0] = \frac{(p_0 - \epsilon)\mu_0 + \epsilon\mu_1}{p_0} = \mu_0 - \frac{\epsilon}{p_0}(\mu_0 - \mu_1). \quad (3)$$

Similarly, $\mathbb{E}[\hat{R}|\hat{S} = 1] = \mu_1 + \frac{\epsilon}{1-p_0}(\mu_0 - \mu_1)$. The reconstructed fairness gap $\hat{F}_{p_0}$:

$$\hat{F}_{p_0} = (\mu_0 - \mu_1) \left(1 - \frac{\epsilon}{p_0(1-p_0)} \right). \quad (4)$$

Thus, the influence of reconstruction error is bounded by

$$|F - \hat{F}_{p_0}| \leq |(\mu_0 - \mu_1)| \frac{\epsilon}{p_0(1 - p_0)} \leq \frac{\epsilon}{p_0(1 - p_0)}, \quad (5)$$

since $|(\mu_0 - \mu_1)| \leq 1$ under Assumption 1 and Assumption 2. This yields $C_\epsilon(p_0)$.

3.2 Concentration Bound

Lemma 1 (McDiarmid's Inequality). *Let $f : \mathcal{A}^N \rightarrow \mathbb{R}$ satisfy the bounded difference property: if changing any single component of the input dataset changes the function value by no more than c_i , i.e.:*

$$\sup_{z_1, \dots, z_N, z'_i} |f(z_1, \dots, z_i, \dots, z_N) - f(z_1, \dots, z'_i, \dots, z_N)| \leq c_i. \quad (6)$$

Then for any $\tau > 0$:

$$P(f(\mathcal{A}) - \mathbb{E}[f(\mathcal{A})] \geq \tau) \leq \exp\left(\frac{-2\tau^2}{\sum_{i=1}^N c_i^2}\right). \quad (7)$$

We apply McDiarmid's Inequality. Consider the function $f(\mathcal{A}) = L_{\tilde{p}_0}$. The sensitivity is maximized when a sample migrates from the smallest group (size N_{min}) to the largest group. The sensitivity c_i for changing one sample is bounded by:

$$c_s = \frac{1}{\lceil \delta N \rceil - 1} + \frac{1}{N - \lceil \delta N \rceil + 1}. \quad (8)$$

The sum of squared differences is $\sum c_i^2 = N \cdot c_s^2$. Applying the Union Bound over $|\mathcal{P}|$ priors for a confidence level η , we solve for the deviation τ :

$$2\tau^2 = N \cdot c_s^2 \cdot \ln\left(\frac{|\mathcal{P}|}{\eta}\right) \implies \tau = c_s \sqrt{\frac{N \ln(|\mathcal{P}|/\eta)}{2}}. \quad (9)$$

This establishes the term:

$$\mathcal{M}(N, \eta, \delta) = \left(\frac{1}{\lceil \delta N \rceil - 1} + \frac{1}{N - \lceil \delta N \rceil + 1}\right) \sqrt{\frac{N \ln(|\mathcal{P}|/\eta)}{2}}. \quad (10)$$

3.3 Robustness

For the robust optimization objective $L_F = \beta \log(\sum \exp(L_{\tilde{p}_0}/\beta))$, by the property of the Log-Sum-Exp function, we have:

$$L_{\tilde{p}_0^*} \leq \beta \log(\sum \exp(L_{\tilde{p}_0}/\beta)) = L_F, \quad (11)$$

for any $\tilde{p}_0^* \in \mathcal{P}$.

3.4 Synthesis of the Bound

Theorem 1 follows directly from chaining the derived inequalities. By the triangle inequality, we decompose the total error as $|F| \leq |\hat{F}_{p_0}| + |F - \hat{F}_{p_0}|$. Substituting Eq. (5) for the reconstruction error, Eq. (9) for the concentration margin, and noting that $L_{\tilde{p}_0} \leq L_F$ (Eq. (11)), we obtain the final generalization bound: $|F| \leq L_F + \mathcal{M}(N, \eta, \delta) + C_\epsilon(p_0)$. $\square$